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Boolean oracle max error: 3.104e-03
Samples used: 100,000
```

The algorithm looked at 100,000 samples, compressed a sketch of the data using a quantum oracle, and the approximation says it's only off by about 0.3%.

```
General-vector l2 error: 7.460e-03
Output state norm: 1.254e-01
Samples used: 60,000
```

The quantum algorithm compressed 60,000 streaming data points into a quantum sketch; the output signal was about 12.5% strength (expected for this type of sketch), and the approximation error was only 0.746%

- meaning it captured the essential structure of the data very accurately, using far less memory than storing everything would require.
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```
Matrix shape: (24, 24), nnz: 111
Matrix element-oracle max error: 6.884e-05
Samples used: 100,000
```

The algorithm took a 24×24 matrix (mostly zeros, 111 real values) and reconstructed it from 100,000 streaming samples using a quantum oracle sketch. The result was nearly perfect — only 0.007% off from the true matrix. This is especially impressive because the algorithm never stored the full matrix; it just sketched it on the fly.

This is the most powerful demo of the three — showing QOS can reconstruct structured data like matrices from streams with remarkable accuracy.

```
Fitted exponent beta: 0.9780
Fitted prefactor C: 2.4725e+02
Log-space R^2: 0.9999
M= 2000 -> error=1.439e-01, fit=1.461e-01
M= 5000 -> error=6.028e-02, fit=5.962e-02
M= 10000 -> error=3.061e-02, fit=3.027e-02
M= 20000 -> error=1.542e-02, fit=1.536e-02
M= 50000 -> error=6.199e-03, fit=6.271e-03
```

As more data samples are added to the quantum sketch, the error keeps shrinking predictably and consistently. Going from 2,000 to 50,000 samples reduced the error by 23x. The theoretical model predicts this behavior almost perfectly ($R^2=0.9999$), which confirms that QOS is a mathematically well-behaved algorithm